AM1 Workplace Project:

Enhancing Clinical Trial Feasibility Through AI: Predicting Trial Durations with Machine Learning

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ST0763 Artificial Intelligence (AI) Data Specialist Level 7

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Contents

List of figures

List of tables

List of Listings

Abstract

Background Accurately estimating clinical trial durations is a critical component of feasibility planning and resource allocation in drug development. Current practices typically rely on simplistic historical averages, which lack standardisation and fail to account for trial-specific variability—such as complexity, geographic distribution, and site capacity. In this project, we explored whether a machine learning (ML)-driven approach could offer a more scalable, data-driven alternative for predicting trial durations, with the potential to support more informed feasibility assessments and strategic clinical operations planning.

Methods An iterative ML development process was undertaken, guided by internal subject matter expert (SME). A curated dataset sourced from ClinicalTrials.gov and enriched via the AACT database was segmented by trial phase (Phase 1 healthy volunteer (P1HV), Phase 1 (P1), Phase 2 (P2), Phase 3 (P3)). Eight non-linear models were selected to capture hypothesised non-linear relationships, including ensemble methods (XGBoost, CatBoost, LightGBM, random forest regressor), a multi-layer perceptron regressor, and simpler models (Decision Tree Regressor, Support Vector Regressor, and K-Nearest Neighbours). A train-train-validation split was applied, and models were optimised using RandomizedSearchCV. Performance was assessed using Root Mean Squared Error (RMSE) and R^2 metrics, using DTR as the baseline for comparison.

Results Ensemble methods—particularly gradient-boosted models—consistently outperformed the baseline DTR and other non-linear models across all phases. Improvements in RMSE ranged from approximately one week in early-phase trials (P1HV, P1) to one month in latter-phase trials (P2, P3). For P1HV, the MLP model achieved the lowest RMSE (90.4 days), while CatBoost, LightGBM, and XGBoost yielded the best results for P1 (110.2 days), P2 (358.3 days), and P3 (401.5 days), respectively. Nonetheless, R^2 scores across all models remained below 50%, and only the P1HV model marginally met its predefined RMSE success threshold ($\leq$90 days). Limited predictive accuracies were attributed to high variability in the dataset, as supported by exploratory data analysis (EDA).

Discussion EDA revealed large standard deviations in trial duration across all phases, reflecting inherent data variability that likely limited model performance. No meaningful clusters were identified through k-means clustering, and outlier removal via phase-specific filtering only modestly reduced spread. Broad, phase-agnostic patterns were observed across therapeutic areas and regions, including long and more variable durations in oncology trials as well as higher P1 activity in Australia, which were in line with SME insights.

Lack of diverse data sources reflecting key factors mediating trial duration such as site capacity, recruitment competition is a key limitation in the project. Future directions include integrating real-time data, enhancing feature engineering, and exploring advanced architectures such as transfer learning and HINT-based models. Model validation against current estimating methods and external models also merits consideration. Planned stakeholder feedback cycles are expected to guide future iterations and support deployment-readiness.

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This project demonstrates the potential of AI/ML to address inefficiencies in clinical trial design and lays the groundwork for more adaptive, data-enabled trial planning strategies across the industry.

Abbreviations

AACT, Aggregate Analysis of ClinicalTrials.gov; AI, artificial intelligence; ANZCTR, Australian New Zealand Clinical Trials Registry; ATMP, advanced therapy medicinal products; CatBoost, categorical boosting regressor; CGT, cell and gene therapy; CRO, clinical research organisation; DTR, decision tree regressor; ElasticNet, ElasticNet regression; EudraCT, European Union Drug Regulating Authorities Clinical Trials Database; HV, healthy volunteers; ICH, International Council for Harmonisation; KNN, K-nearest neighbours regressor; KPI, key performance indicator; LGBM, light gradient boosting machine regressor; MAD, multiple ascending dose; MeSH, medical subject headings; ML, machine learning, MLP, multi-layer perceptron regressor; P1, Phase 1; P2, Phase 2; P3, Phase 3; P4, Phase 4; RFR, random forest regressor; RMSE, root mean squared error; SAD, single ascending dose; SME, subject matter expert; SVR, support vector regressor; XGBR, XGBoost regressor.

1 Introduction and Background

1.1 Background

1.1.1 Clinical trials and drug development

Clinical trials are an indispensable part of drug development to enable preclinical scientific discoveries in the laboratory to be tested in humans through clinical development before reaching the market as a regulatory-approved treatment for a disease. In clinical development, drug developers assess various aspects of a drug candidate to understand its properties and activity in humans. These properties include a candidate's efficacy (i.e. whether the drug elicits the intended therapeutic effect in the body); its safety or tolerability (i.e. whether the drug is safe and well tolerated in humans); as well as its pharmacology (i.e. how the drug is absorbed, distributed, metabolised, and excreted, and how its therapeutic effects are exerted and maintained over time). To ascertain this, trials are designed to test drugs in both healthy volunteers (HV) and in patients across three main phases, namely:

- Phase 1 (P1): Primarily assesses the safety and tolerability of a drug candidate, typically in healthy volunteers (HV). However, for indications such as oncology, where treatments are often cytotoxic, trials are conducted directly in patients.
- Phase 2 (P2): Evaluates efficacy and continues safety assessments in a small patient population.
- Phase 3 (P3): Confirms efficacy and safety in a larger, more diverse patient group to support regulatory approval.

Within Phase 1, further subcategorisation by dosing strategy exists, such as Single Ascending Dose (SAD) or Multiple Ascending Dose (MAD) studies, often termed as Phase 1a and Phase 1b, respectively.

While these phases offer a general framework, actual trials often deviate from these broad features. For instance, P1 studies may include preliminary efficacy endpoints beyond testing for safety and tolerability, and the number of patients enrolled in P2/P3 studies can vary substantially, especially in rare diseases such as Duchenne Muscular Dystrophy or sickle cell disease, where trials may enroll fewer than 100 patients. Hybrid designs, such as P1/2 or P2/3, are also increasingly common, particularly in areas with limited patient populations.

More recently, adaptive trial designs, which allow modifications based on interim analyses, have also gained traction. These novel designs enable changes to sample sizes, treatment arms, as well as early trial termination based on predefined criteria, allowing for more flexible and efficient trial execution.

Despite scientific and methodological advances, clinical development remains a high-risk, high-cost endeavour, often requiring over a decade and up to $2 billion to bring a single drug to market. These challenges underscore the urgent need for more efficient and predictive tools to optimise trial planning and execution.

1.1.2 AI and ML in biopharma

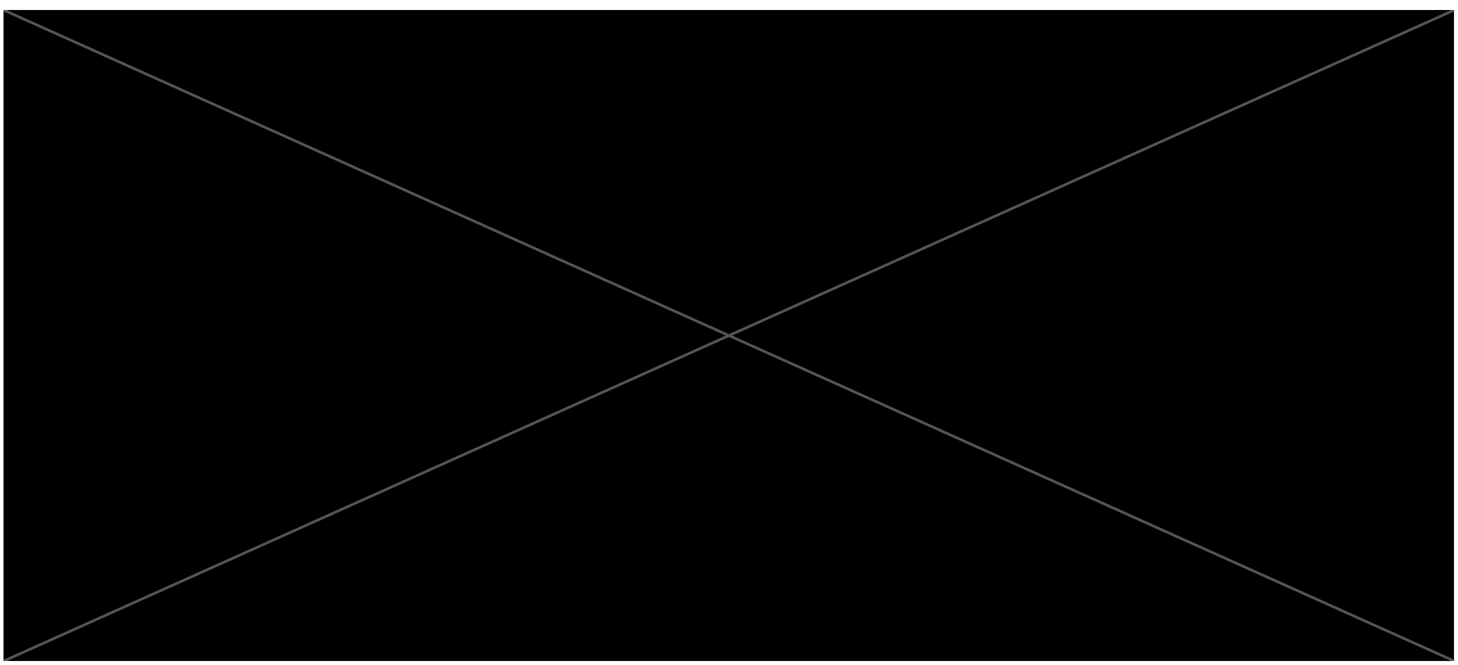
Artificial Intelligence (AI) and Machine Learning (ML) have recently emerged as transformative technologies in biopharmaceutical R&D. Their most prominent applications have been in drug discovery, such as in protein structure prediction via DeepMind's AlphaFold, which won the Breakthrough of the Year accolade and has now progressed to AlphaFold 3.

A number of AI-discovered or AI-optimized drug candidates, such as those from Insilico Medicine, are now in P2 clinical trials (Insilico Medicine, 2023), validating the field's potential. AI-driven approaches such as lab-in-the-loop systems are also increasingly adopted to accelerate early-stage experiments.

Despite these advancements, clinical development remains a critical and irreplaceable phase to evaluate drug candidates in real human populations, as mandated by international regulatory frameworks such as the ICH E6(R2) Good Clinical Practice (GCP) guidelines (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), 2016).

The complexity of modern trials—driven by increasing numbers of endpoints, procedures, eligibility criteria, and geographic expansion—has only intensified. As highlighted by Bozzetti et al., 2025; Wu et al., 2022, the clinical trial process has become more demanding, not only in design but also in execution. A persistent challenge remains the high attrition rate across all phases of development, often attributed to scientific, technical, operational, and commercial issues.

While AI/ML have been applied extensively to scientific derisking, such as target validation and compound screening, their use in improving operational efficiency – especially in trial planning and recruitment – remains underexplored. Given their ability to process high-dimensional data and generate actionable predictions, AI/ML tools hold strong potential to transform how trials are designed and executed.



1.1.4 Business problem

One of the approximation methods of estimating recruitment rates is calculating the proportion of patients and numbers of centres relative to the length of the trial in months. This does not account for other trial-specific factors that mediate recruitment including:

- Trial complexity (e.g. Burden of assessments)

- Dynamically changing patient pool size (e.g. due to region-specific competition from ongoing trials, and emerging curative treatments)
- Real-time capacity of trial sites
- Intrinsic patient pool size from disease prevalence per region

This may result in inaccuracies in trial duration predictions and lower the quality of feasibility assessments and clinical operation recommendations we provide our clients. More accurately predicting duration also leads to better resource allocation, financial planning, risk management.

More accurate predictions of trial duration could accelerate data gathering exercises for feasibility studies and clinical operations projects, better inform patient recruitment strategy in trial design and risk management.

This will enable more refined trial cost estimates, ultimately derisking and driving better outcomes for our clients' clinical development programmes. The project is also in line with broader digital transformation initiative of shifting towards an increasingly automated, data-driven decision-making culture within the organisation.

1.2 Project Objectives, Scope, and Success Criteria

The aims of this project were to build a trial duration prediction tool, as a proxy measure of recruitment rate to improve on recruitment rate estimation methods which involve heavily manual, static, and unscalable processes. As a standalone trial prediction support tool, this has scope for further upscaling and incorporation into a potential multimodal data product in the long term, such as an AI-assisted clinical development planning support tool as proposed by Jung et al., 2022 in the long term.

This project also set out to be conducted with AGILE methodologies, known as software development best practices, which has been increasingly adopted in ML projects (Siebert et al., 2019). This is due to its benefits of encouraging collaborative practices and continuous feedback and accelerated development to reach a minimal viable product (MVP) through short and multiple iterative cycles of feedback incorporation and development.

Motivated by the AGILE methodology, the business requirements and success criteria have been identified and defined through regular discussion with internal subject matter experts (SMEs), Head of Clinical Operations, Geoff Brooks. This ensures a quantitative measure of progress as well as the quality of the project outputs. The following shows a phase-specific "acceptable margins of error (in days)" with root mean squared error (RMSE) as the prioritised evaluation metric for predictions derived from the ML tool:

Dataset	Ideal RMSE margin in trial duration predictions
P1/P1HV	$\leq$90 days (a quarter)
P2/P3	$\leq$180 days (half a year)

Table 1: Defined criteria of success in terms of RMSE margin of trial duration predictions per phase, guided by internal SMEs.

2 Methods

Key stages in a standard ML project workflow were carried out in the project, starting with the collection of data from the public clinicaltrials.gov database enriched by data from AACT. This is followed by a series of data preprocessing and filtering steps. Subsequently, four phase-specific Datasets were created and prepared for model development. Exploratory data analysis (EDA) including k-means clustering was carried out throughout to guide data preprocessing decisions. After preprocessing, data was prepared by train-test-validation split for training with a selected set of mainly non-linear models, from decision tree regressor to more complex ensemble methods. Models were tuned using RandomizedSearchCV and performances were evaluated by two regression metrics (root mean squared error and R^2 score). AGILE methodology was employed throughout the project, allowing for iterative development cycles and continuous feedback from subject matter experts (SMEs).

Fig 1 below shows the key steps involved.

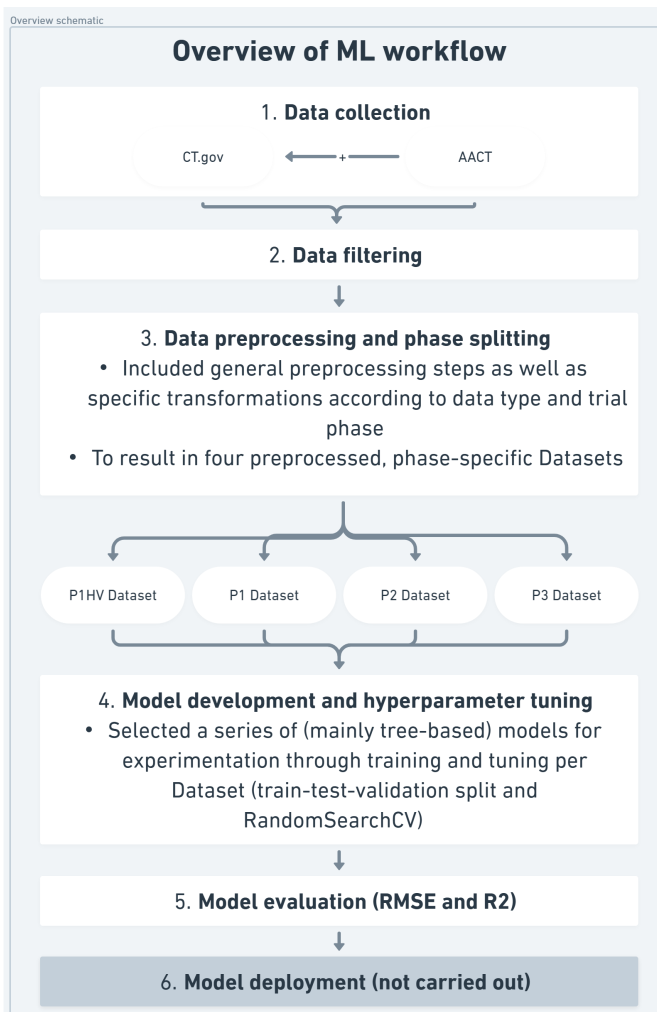


Figure 1. Overall schematic of ML workflow Key stages in the ML workflow, including data collection, preprocessing and filtering, and subsequently model development, tuning, evaluation, and deployment, which was not carried out in this project

2.1 Data Selection, Collection and Preprocessing

2.1.1 Data Selection and Collection

I sourced the dataset from the prominent US-based trial registry database covering clinical studies conducted across the globe at https://clinicaltrials.gov. This was in the form of a static CSV with 30 features and 507744 rows when accessed on 4th September 2024. This was subsequently enriched by the additional structured data from the relational Aggregate Analysis of ClinicalTrials.gov (AACT) Database (CCTI, n.d.). PostgreSQL was used to access a list of selected relevant features upon manual inspection, which was merged to form the starting dataset using the unique trial identifier.

2.1.2 Data Filtering and Preprocessing

To build a model for predicting clinical trial durations, a series of data preprocessing steps were undertaken, as follows:

I began by filtering out non-interventional and incomplete studies, and further restricted the dataset to industry-funded trials with conventional pharmacological interventions, specifically those labelled as `DRUG` or `BIOLOGICAL`. Trials with missing start or completion dates were excluded, along with those lacking values in key structural variables such as phase, location, or enrolment. Data types were standardised, including parsing date fields into `datetime` objects and converting trial phase information into categorical variables.

Missing values for numerical features were imputed with the median. I also removed outliers defined by z-score greater than 3, corresponding to data points exceeding a threshold of three standard deviations (SD) assuming normal distribution.

New features were created through feature engineering, such as a binary indicator for HV-only studies. In particular, P1 subcategories were engineered by applying text parsing to extract information on (A) trial subtypes and (B) dosing design. Specifically, text were parsed from the `Study Title`, `Brief Summary`, and `Primary Outcome Measures` fields to match for the following text strings specifically:

- (A) efficacy, safety/tolerability, and pharmacology, and
- (B) single ascending dose (SAD) or multiple ascending dose (MAD).

See Listings 2 (A) and 3 (B) respectively for code snippets.

To support therapeutic area stratification, I mapped trial conditions to standardised therapeutic area using a custom-made MeSH-based dictionary (Listing 4). This disctionary was constructed based on the publicly available MeSH dictionary developed by the National Library of Medicine to organise biomedical information (of Medicine, n.d.). Geographic location data were similarly mapped to broader global regions using a keyword-based regional dictionary (Listing 5).

Finally, I applied one-hot encoding to the therapeutic area and region variables. Redundant or low-information columns were removed based on domain knowledge and relevance to duration prediction.

The target variable, `Trial_duration_Primary_Completion`, was derived from date differences between the "Start Date" and the "Primary Completion Date" of each trial. As a proxy to recruitment duration, trial duration to reach Completion Date versus and the Primary Completion Date was not significantly different, which was also confirmed through visualisation and inspection, therefore time to Primary Completion was selected to create the target variable, and shall be referred to interchangeably with the word "trial duration" henceforth. As part of cleaning, trials with duration of 0 days were excluded. Subsequently, after phase splitting, outliers were further removed with the SME-guided upper limit of standard trial duration specific to each phase (Table 2.

Dataset	**Upper limit of accepted trial duration (days)**
P1/P1HV	600
P2	2200
P3	3000

Table 2: Define upper limit threshold of accepted trial duration (days) per trial phase to identify outliers for exclusion, guided by internal SMEs.

Table 3 below shows the lists of trial features in the dataset before and after preprocessing.

Preprocessing steps are also summarised in Figure 2.

Before Preprocessing (N = 507,744)	After Preprocessing (N = 42,871)
NCT Number	Phases
Study Title	Enrollment
Study URL	total_primary_outcomes
Acronym	total_secondary_outcomes
Study Status	number_of_arms
Brief Summary	has_expanded_access
Study Results	Location (site count)
Conditions	Drug_Type
Interventions	primary_completion_year
Primary Outcome Measures	final_P1_category
Secondary Outcome Measures	final_sad_mad_category
Other Outcome Measures	Trial_duration_Primary_Completion
Sponsor	Trial_duration_Completion
Collaborators	Conditions_Contains_HV
Sex	Age_Contains_CHILD
Age	Collaborators_IsNaN
Phases	Allocation
Enrollment	Intervention_Model
Funder Type	Masking
Study Type	Primary_Purpose
Study Design	*Infectious Diseases*
Other IDs	*Oncology/Solid Tumours*
Start Date	*Rheumatology*
Primary Completion Date	*Gastroenterology*
Completion Date	*Stomatognathic Diseases*
First Posted	*Respiratory*
Results First Posted	*Otorhinolaryngology*
Last Update Posted	*CNS/Neurology*
Locations	*Ophthalmology*
Study Documents	*Urology/Male*
	Urology/Female Health
	Cardiovascular
	Haematology
	Congential/Hereditary/Rare Diseases
	Dermatology/Connective Tissue Diseases
	Metabolic
	Endocrinology
	Immunology
	Pathological Conditions, Signs and Symptoms
	Chemically-Induced Disorders
	Wounds and Injuries
	Other
	_US_
	North America (excluding US)
	Europe
	Asia
	Australia/Oceania
	Latin America
	MENA
	Sub-Saharan Africa
	ROW
Number of Features: 30	**Number of Features: 52**

Table 3: List of features before and after data preprocessing. One-hot encoded categorical features are highlighted in particular, where therapeutic area features are marked with *asterisks* and geographic region features in *italic*.

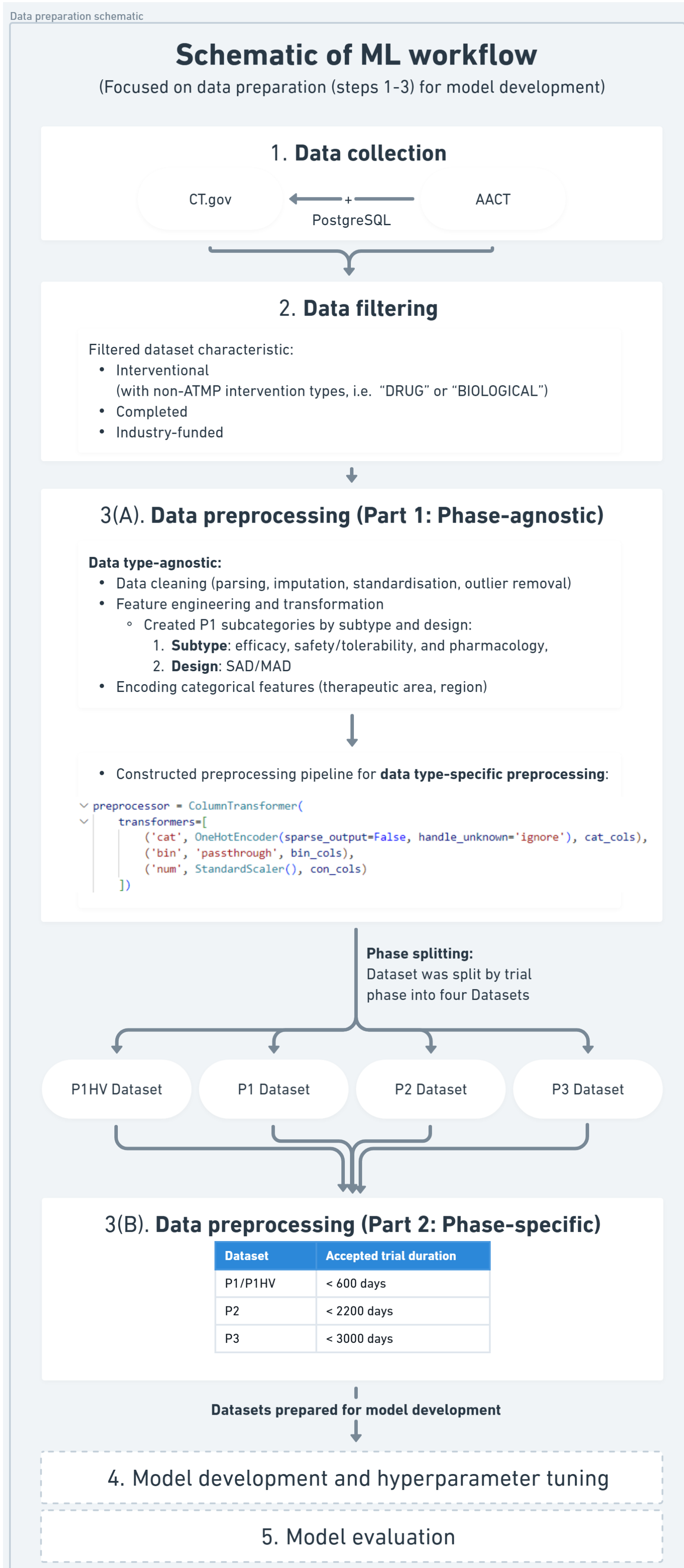


Figure 2. Workflow schematic focused on data preprocessing steps

After preprocessing, the dataset was divided into 4 phase-specific Datasets (Table 4): Phase 1 healthy volunteers (HV)-only trials (P1HV), Phase 1 (P1), Phase 2 (P2), and Phase 3 (P3). The decision to divide up the dataset for phase-specific model development was guided by SME and domain expertise, prioritising the main trial phases, as each phase is characterised by phase-specific distinguishing features, they are considered as distinct entities. The P1HV and P1 (patient-only) Datasets were separated to isolate the trials that only recruit HV, as this has implications on recruitment speed and hence duration.

Phase	Description
P1HV	Industry-funded, P1 (HV only), interventional, drugs/biologics trials
P1	Industry-funded, P1 (patients only), interventional, drugs/biologics trials
P2	Industry-funded, P2 (patients only), interventional, drugs/biologics trials
P3	Industry-funded, P3 (patients only),interventional, drugs/biologics trials

Table 4: Key description of the resultant four phase-specific Datasets that were split from the original trial dataset after the phase-agnostic data preprocessing steps.

The preprocessed dataset before filtering by phase had 42871 studies and 52 features (Table 3; after dividing the data by trial phase and applied phase-specific preprocessing, the four final phase-specific Datasets included 5483 P1HV, 7216 P1, 10312 P2, and 10205 P3 trials (Table 5).

Exploratory Data Analysis (EDA) Exploratory Data Analysis (EDA) was performed through the data preparation stage which helped guide preprocessing decisions. Aspects examined at EDA included feature distributions, temporal patterns, and variability across phases and therapeutic areas. Data spread of features and the label were visualised using libraries such as matplotlib and seaborn, summarised by year and by phase. Descriptive statistics were also obtained to understand the inherent data distribution of the label (trial duration) per Phase.

Clustering K means clustering was also applied during EDA to explore the possibility of deriving new categorical features by grouping similar trials in the same clusters. These clustering results were examined for potential subsequent downstream feature engineering. Cluster validity was assessed using the elbow method and silhouette scores to evaluate the quality of the formed clusters in terms of their compactness and the degree of separation.

2.2 Model Development and Optimisation

After preparing the data for modelling, train-test-validation split was carried out first with a 80:20 split to create a hold-out unseen test set (20%), followed by a 80:20 split on the remaining 80% to result in the final training (64%), validation (16%), and test (20%) sets with the scikit-learn library. Feature preprocessing was handled via a `ColumnTransformer` to carry out transformations specific to the data type of the features (categorical, continuous, or binary).

Model Selection

Reviewing similar work in the domain from the literature (Long et al., 2023; Wu et al., 2022; Yue et al., 2024), a series of models were selected based on the hypothesis of that the feature relationships in the trial data are largely non-linear, such as the non-linear relationship between trial duration and the number of countries/geographical regions involved as reflected in the work of Wu et al., 2022. A simple Multi-Layer Perceptron (MLP) neural network was also implemented to explore deep learning approaches. Therefore, the following regression algorithms were explored:

- **Tree-Based Models:** Decision Tree Regressor (DTR) (as baseline), Random Forest Regressor (RFR), XGBoost Regressor (XGBR), LightGBM Regressor (LGBM), CatBoost Regressor (CatBoost)
- **Other Models:** Support Vector Regressor (SVR), K-Nearest Neighbors (KNN), Multi-Layer Perceptron (MLP)

Hyperparameter Tuning

Hyperparameter tuning was performed using `RandomizedSearchCV` with 5-fold cross-validation for all non-baseline models. Each tuning run was repeated across multiple random seeds (`random_state` = 42, 0, 1, 2, 3) to ensure result robustness. The models were retrained on the full training dataset and evaluated on the models were evaluated on their performance on the held-out test set.

RandomizedSearchCV was used alongside learning curves to track performance and detect any overfitting or data scarcity issues.

2.3 Model Evaluation and Validation

Model performance was evaluated on the hold-out test set in an iterative manner, such as after feature creation, to guide preprocessing decisions. Two standard regression metrics were used: Root Mean Square Error (RMSE) and the R^2 score. RMSE was prioritised as the primary evaluation metric due to its interpretability as it is expressed in the same unit as the target variable. This is a more intuitive measure that is conducive for effectively communicating results with non-technical stakeholders in the wider team, whilst retaining the advantage of penalising large errors compared to mean absolute error. On the other hand, R^2 score represents the proportion of variance in the target variable explained by the model to provide a relative measure of model performance.

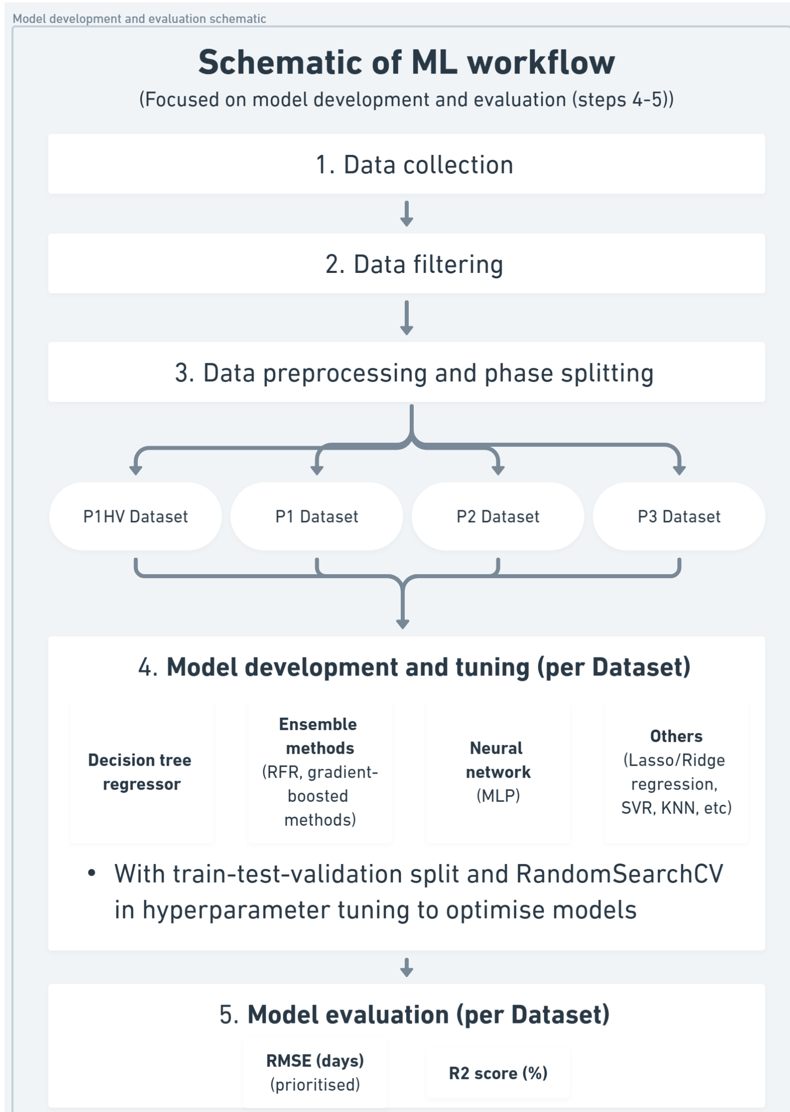


Figure 3. Workflow schematic focused on model development, tuning, and evaluation

External validation of the model has not been carried out in this project, such as the collection of a lung cancer dataset as a means of external validation for the P1 lymphoma model in the work of Long et al., 2023. The project also did not reach deployment stage. Nevertheless, I assessed predictions on the test set by box plot visualisations using seaborn of the predicted versus actual trial durations of the best-performing model for each phase-specific Dataset across therapeutic areas (Fig 5).

Project Management An iterative and collaborative approach was adopted throughout the project, motivated by the AGILE methodology which was widely considered as best practice for standard software development. For instance, I conducted regular meetings with data science colleagues and internal non-technical SMEs to obtain feedback to be incorporated in an iterative manner, contributing to various aspects of the project including feature creation and prioritisation, defining success criteria, as well as general domain knowledge sharing.

3 Results

Amongst the eight non-linear models developed and evaluated across the four phase-specific Datasets, the ensemble methods (particularly the gradient-boosted methods) showed the strongest predictive performances based on RMSE. Compared to the baseline decision tree regressor (DTR), the best-performing models delivered an observed improvement of approximately one week in early-phase trials (P1HV, P1) and a month in latter-phase trials (P2, P3). However, the R^2 scores for all models remain below 50%, highlighting a need for further improvement. Models have also not been validated against current averaging trial prediction method or external models. In terms of meeting the predefined success criteria, the outputs marginally achieved success for P1HV but fall short for P1, P2, and P3, which may be attributed to the high variability inherent in the dataset.

3.1 EDA findings

Exploratory data analysis (EDA) enabled a better understanding of the dataset. Descriptive statistical analysis of of the trial duration label revealed a high inherent variability in the phase-specific Datasets, evidenced by large SD relative to the respective means (Table 5).

Table 5 presents the descriptive statistics of the ground-truth trial duration label for the four phase-specific Datasets before and after preprocessing:

Statistic	P1HV	P1	P2	P3
Before Filtering				
N	5667	10353	10524	10309
Mean	158.5	505.0	657.3	725.7
Median	92.0	288.0	517.0	578.0
SD	176.9	556.8	544.8	603.8
Min	1	1	3	2
Max	2160	5660	6392	6877
After Filtering (Upper Duration Limit Applied)				
N	5483	7216	10312	10205
Mean	135.7	208.2	610.9	693.4
Median	92.0	162.0	504.5	577.0
SD	120.3	158.1	431.6	505.9
Min	1	1	3	2
Max	596	599	2197	2953

Table 5: Descriptive statistics of the trial duration label and sample sizes for each preprocessed phase-specific Dataset before and after removing outliers using phase-specific upper duration limits, revealing high inherent variability in Datasets evidenced by relatively large SDs.

Additionally, k means clustering was applied during EDA intended to explore potential clusters in the higher-dimensional feature space for a possibility of incoporating it as a new cluster-based feature or guide further preprocessing steps such as outlier removal. However, there were no apparent clusters across the Datasets when applying clustering both before and after phase-filtering when evaluated using the elbow method across multiple values of k.

3.2 Findings from Model Experimentation

A prioritised set of eight regression models were trained and tuned to be assessed by their trial duration predictions across trial phases. Model performances were evaluated using the prioritised RMSE metric with the aforementioned justification, and compared against the baseline DTR. Overall, ensemble-based techniques demonstrated stronger predictive performances.

Detailed model performances (in RMSE, days) across the four Datasets are as follows:

Model	P1HV	P1	P2	P3
DTR	*98.8*	*121.6*	*395.9*	*449.9*
SVR	93.7	114.8	378.2	429.4
KNN	93.5	116.6	376.5	416.8
RFR	91.6	112.1	367.7	411.6
XGBR	**90.4 (-8.4)**	110.3	360.1	**401.5 (-48.4)**
LGBM	**90.4 (-8.4)**	111.1	**358.3 (-37.6)**	407.3
CatBoost	90.5	**110.2 (-11.4)**	360.5	404.5
MLP	**90.4 (-8.4)**	113.9	361.8	415.7

Table 6: Root Mean Squared Error (RMSE, in days) for each model across the four phase-specific Datasets. Bolded blue values indicate the best-performing model(s) for each Dataset, with the improvement from the baseline DTR RMSE shown in parentheses. Red italicized values indicate the worst-performing model across all Datasets (also the baseline DTR).

Table 7 below shows a summarised overview of the best-performing models (determined by the prioritised RMSE metric) across the phase-specific Datasets:

Dataset	Best Model	Best Model RMSE	Best Model R^2
P1HV	**MLP**	90.4 days	43.1%
P1	**CatBoost**	110.2 days	43.2%
P2	**LGBM**	358.3 days	33.0%
P3	**XGBR**	401.5 days	35.5%

Table 7: The best model performances in both evaluation metrics (RMSE and R^2) from the best-performing models for each Dataset.

P1HV	
Model (Seed)	**Optimal Parameters**
XGBR (1)	subsample: 0.8; reg_lambda: 1; reg_alpha: 0; n_estimators: 100; max_depth: 3; learning_rate: 0.1; gamma: 1; colsample_bytree: 0.7
LGBM (42)	subsample: 0.9; n_estimators: 100; max_depth: 3; learning_rate: 0.0733; colsample_bytree: 0.8
MLP (42)	solver: adam; learning_rate: adaptive; hidden_layer_sizes: (50,); alpha: 0.01; activation: tanh
P1	
Model (Seed)	**Optimal Parameters**
CatBoost (42)	learning_rate: 0.1; l2_leaf_reg: 1; iterations: 300; depth: 8
P2	
Model (Seed)	**Optimal Parameters**
LGBM (1)	subsample: 0.9; n_estimators: 150; max_depth: 8; learning_rate: 0.0522; colsample_bytree: 0.7
P3	
Model (Seed)	**Optimal Parameters**
XGBR (3)	subsample: 0.8; reg_lambda: 1; reg_alpha: 10; n_estimators: 500; max_depth: 10; learning_rate: 0.01; gamma: 1; colsample_bytree: 0.5

Table 8: Optimised hyperparameters for the best-performing models per phase-specific Dataset. Seed values for model reproducibility are shown in brackets.

On average, the more complex algorithms such as ensemble methods outperformed baseline DTR as well as the rest of the models, with gradient-boosted methods (XGBR, LGBM, and CatBoost) delivering slightly better outcomes than RFR with the Random Forest Regressor (RFR) with the bagging approach. Notably, XGBR achieved the lowest RMSE for both P1HV (90.4 days) and for P3 (401.5 days). For P1HV in particular, both MLP and the other gradient-boosted methods also achieved similarly strong predictive performances, with RMSE of 90.5 days for CatBoost and 90.4 days for MLP and LGBM as well.

For P1, CatBoost had the highest predictive accuracy (with RMSE of 110.2 days), while LGBM outperformed all other models for P2 (358.3 days), equivalent to approximately a year's worth of error from the ground-truth duration value.

In terms of improvement from the baseline, the best-performing models such as XGBR, LGBM, and CatBoost reduced the RMSE by approximately 8 to 11 days (roughly equivalent to a week's worth of improvement) compared to baseline DTR with RMSE of 98.8 days (P1HV) and 121.6 days (P1). For latter-phase trials (P2 and P3), these models improved the RMSE by about 38 to 48 days, corresponding to roughly a month's worth of improvement, with LGBM reaching

358.3 days (vs 395.9 days for DTR) in P2 and XGBR achieving 401.5 days (vs. 449.9 days) in P3 (Table 6).

The conclusions drawn from using RMSE as the evaluation metric are coherent with those reflected from the R^2 scores, wherein the best models for P1HV and P1 both achieved R^2 scores of approximately 40%, higher than the best P2 and P3 models both achieving approximately 28%. Unfortunately, no model across phases achieved R^2 of over 50%, which is consistent with and may reflect the previously highlighted potential high inherent variability and noise in the Datasets (Table 5.

Figure 4 shows the evaluation metrics of the best models against the spread of data in the original phase-specific Datasets, where the latter-phase trials (P2, P3) have larger absolute values of SD (as in 5) than earlier ones (P1, P1HV), demonstrating the potential hindering effect of the inherent variability in the Datasets on the models achieving more robust predictive performances.

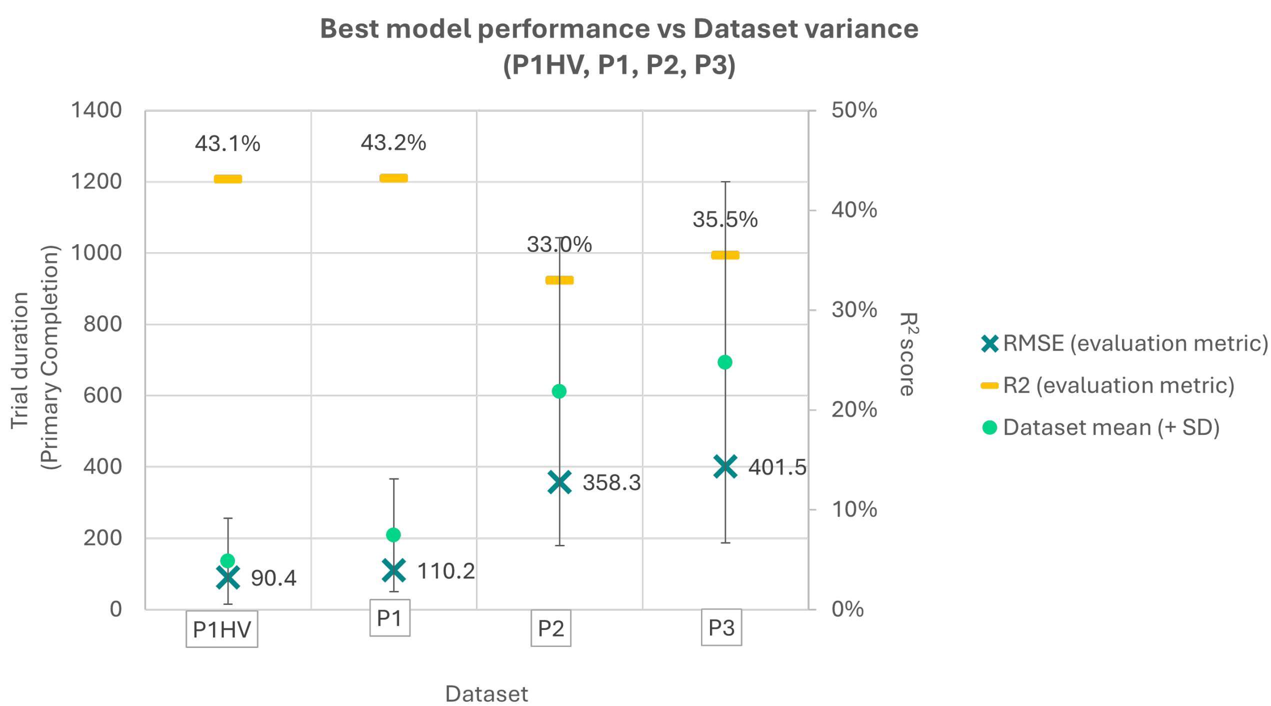
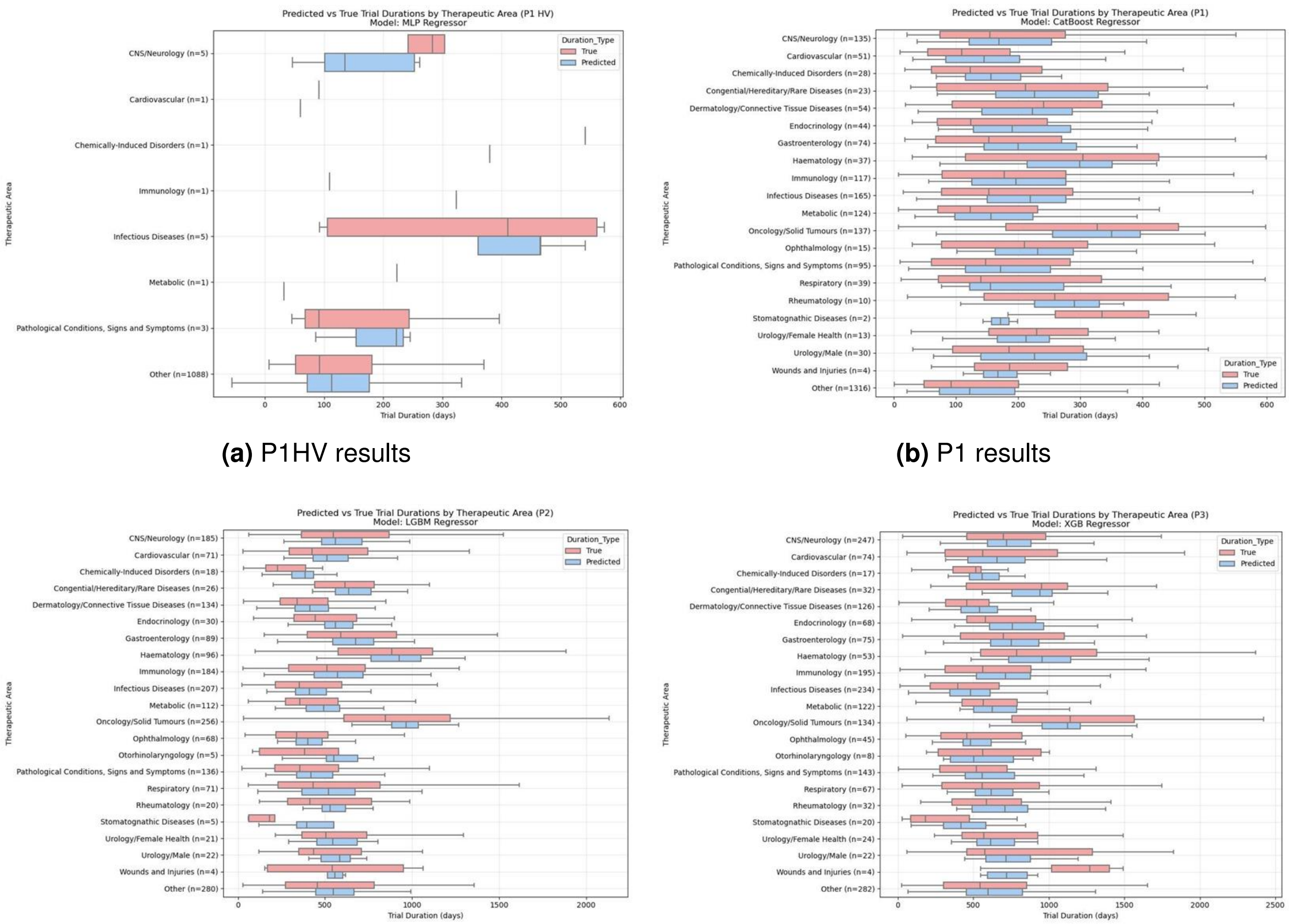


Figure 4. Summary of best model performances per Dataset Summary of performances of the best-performing models in RMSE (days) and R^2 score (%) against the original variance in the phase-specific Datasets.

From visualising the predicted versus actual trial durations of the best-performing model on the test set for each phase across therapeutic areas (Fig 5), although the model-predicted values "tend" towards the ground-truth trial duration values, it is worth noting the high variance within therapeutic area per phase evidenced by the large boxes (corresponding to the inter-quartile range). It was expected that high numbers of P1HV trials were classified as "Other", because their therapeutic area feature should be "healthy volunteer", and would not have been associated with any particular therapeutic area. Across P1, P2, and P3 results, "Oncology/Solid

Tumours" were consistently associated with longer and more variable trial durations, which is in line with the pattern displayed in Supplmentary Figure 1. Furthermore, it is noteworthy from Supplementary Figures 1 and 3 that both features ("Therapeutic Area" and "Region") show similar patterns of spread of trial durations that are shared across trial phases.



(a) P1HV results

(b) P1 results

(c) P2 results

(d) P3 results

Figure 5. True (red) versus predicted values (blue) of trial durations by the best-performing models across therapeutic areas for the test set for each phase-specific Dataset.

On the whole, the P1HV model with RMSE of 90 days has marginally met the predefined success criteria to be considered significantly better than the predictive performance of the current manual process (RMSE $\leq$ 90 for P1HV/P1). However, the best P1, P2, and P3 models did not meet the benchmarks guided by SME, indicating ideal predictive performances of RMSE $\leq$ 90 for P1, as well as RMSE $\leq$ 180 days for P2 and P3 trials (Table 1).

Nevertheless, I have demonstrated the potential of ML in enhancing the process of trial duration predictions. These improvements can support in devising strategies on trial recruitment and other clinical operational decisions. Further scope of improvement in predictive scores through a number of avenues, particularly for the latter-phase trials (P2 and P3), are to be discussed in subsequent sections.

Looking ahead, the collection of stakeholder feedback (which has not been achieved within the duration of this project) for incorporation in subsequent cycles of iterations is anticipated to be conducive to the continual tool development process. This will also confirm, capture, and quantify the degree of efficiency gains hopefully achieved upon tool deployment.

4 Discussion and Implications

Clinical trials are complex and costly, underscoring the need for resource allocation and risk management to enable optimal operational efficiency. Feasibility studies are an essential component of planning in drug development programmes. Enhancing accuracy in clinical trial duration estimation presents a critical opportunity to leverage AI/ML to improve feasibility assessments. Current methods rely on historical averages and heuristic judgments, neglecting trial-specific variables that may mediate recruitment, such as trial complexity, geographic distributions, and site capacity. This project set out to explore whether a ML-based approach could offer a more quicker, more data-driven and scalable alternative for predicting trial durations, with the potential to support informed feasibility assessments and strategic clinical operations recommendations.

The full ClinicalTrials.gov dataset, enriched via the AACT database, was processed and segmented into four phase-specific Datasets (P1, P1HV, P2, P3) based on SME guidance. To capture hypothesised non-linear relationships, a range of non-linear models were chosen for development and evaluation, including ensemble methods – boosting (XGBR, CatBoost and LGBM) and bagging (RFR) – alongside an MLP regressor, as well as other simpler models such as SVR and KNN. Across all phases, the ensemble models, particularly gradient-boosted ones, consistently outperformed and reduced the RMSE (as the prioritised evaluation metric) compared to the baseline DTR. Performance improvements translated to approximately one week for early-phase trials (P1HV, P1) and one month for latter-phase trials (P2, P3).

This performance improvement is expected, as ensemble methods aggregate multiple decision trees, reducing the risk of overfitting and increasing model generalisability. While DTR offers high interpretability, ensemble models may be preferred considering the trade-off of predictive accuracy and model complexity. Nevertheless, it may be important and valuable to understand which features contributed to predictied trial durations to enable manual checking and confirmation by relevant stakeholders. Local explainability techniques such as local SHAP, LIME, or ANCHOR can be applied post hoc to explain individual predictions.

When assessing project output according to predefined success criteria, only P1HV model marginally met the RMSE threshold considered to be a significant improvement to current estimating methods (guided by SME). Indeed, no model yielded R^2 scores above 50%, highlighting limited overall predictive accuracy across models and trial phases. This suggests that key predictive signals remain uncaptured in the current dataset and reinforces the need for additional data sources, such as real-time site capacity data from clinical research organisations (CROs) and trial sites.

Exploratory data analysis (EDA) further underscored the observation of high inherent variability in the data, evidenced by large standard deviations relative to the mean observed across all trial phases. Besides, k means clustering failed to identify clear clusters. Additionally, attempts to mitigate noise through outlier filtering using z-scores still resulted in outliers such as trials with implausibly short durations of 1–10 days (Table 5, which were also not excluded despite lowering the z-score cutoff to 2 or 1.

These findings support the notion that irreducible data noise is a significant limiting factor to the model's predictive performance. This may, in part, explain the baseline DTR's underperformance, due to its sensitivity to noise and tendency to overfit. In contrast, ensemble methods would yield more robust predictions due to its generalisability when combining learning and predictions from a ensemble of decision trees.

One noteworthy insight from the EDA is the consistent, broad patterns observed across therapeutic areas and geographic regions that appeared to be phase-agnostic (see Supplementary Figures 1 and 3). These observations align with SME input and established domain knowledge, such as the high concentration of overall trial activity in the US and Europe, as well as the high frequency of P1 trials in Australia in particular. Additionally, oncology trials were identified as both longer in trial duration and consistently variable across trial phases, which is also consistent with domain knowledge. Further investigations. These findings present a promising direction for future work, particularly in validating and transmitting to the wider team other potentially actionable insights uncovered through the analyses.

4.1 Future Opportunities

There are several avenues worth exploring and investigating in the future.

One potential improvement involves transitioning from the static CSV download to establishing live data links, such as via the ClinicalTrials.gov API, to enable continuous and real-time updates.

While this project focused on non-linear models, linear models (e.g. Lasso, Ridge, and Elastic Net regressors) were briefly explored. These warrant further investigation to test the hypothesis that certain features (such as number of arms, sites, endpoints, or enrollment size) that may exhibit linear relationships with trial duration, particularly due to their correlation with trial complexity.

Additionally, hyperparameter tuning was conducted manually in this project using RandomizedSearchCV across multiple random seeds, with performance metrics and optimal parameter combinations logged to versioned Excel files for traceability. This approach was manual and inefficient. Looking ahead, experiment tracking frameworks such as MLflow could be considered to streamline model development and enable a more automated model experimentation, comparison and logging process.

Beyond technical refinements, there are broader opportunities to improve the project on a higher level. For instance, validation against current estimating methods using historical data or external models (e.g. Long et al.'s P1 lymphoma model, and the Hierarchical Attention Transformer (HINT)-based interpretable trial duration prediction tool by Yue et al.) has not yet been conducted. The former would require further discussion with internal SME to define a use case to test the model.

There is also scope to embed this model within a broader data product incorporating multimodal inputs. For example, integrating operational, disease-, and drug-specific characteristics through more advanced architectures—such as transfer learning or the HINT framework proposed by Yue et al.

Upon deployment, stakeholder feedback will be actively sought to guide future iterations and inform broader product development. This aligns with the AGILE methodology adopted throughout the project, which enabled iterative refinement via sprint cycles. Obtaining SME input has proved particularly valuable in shaping key aspects of the project particularly at the early phases of the project, contributing to aspects such as feature engineering, phase segmentation, and predefined success criteria.

4.2 Caveats and Limitations

There are limitations and potential sources of biases, error, and uncertainty in the project that may merit consideration.

Intrinsic limitations

The models' limited predictive accuracy is partly attributable to the inherent label and feature noise. Irreducible error may arise from persistent outliers, such as the aforementioned trials with abnormally short durations, that could not be excluded even with loosened z-score thresholds. In addition, the mapping of trial conditions to therapeutic areas using a custom MeSH-based dictionary was incomplete. Further uncertainty stems from heterogeneous data inputs, such as variation in sponsor-specific reporting templates.

On a more granular level, key indicators of trial complexity, such as the number of inclusion/exclusion criteria and patient visits, have not been engineered and included in the dataset. These features, which indicate the burden of assessments and are known to increase trial complexity and duration, could provide valuable differentiation among otherwise similar trials. This has also been highlighted by prior work (Wu et al., 2022), and thus warrants further investigation in future iterations.

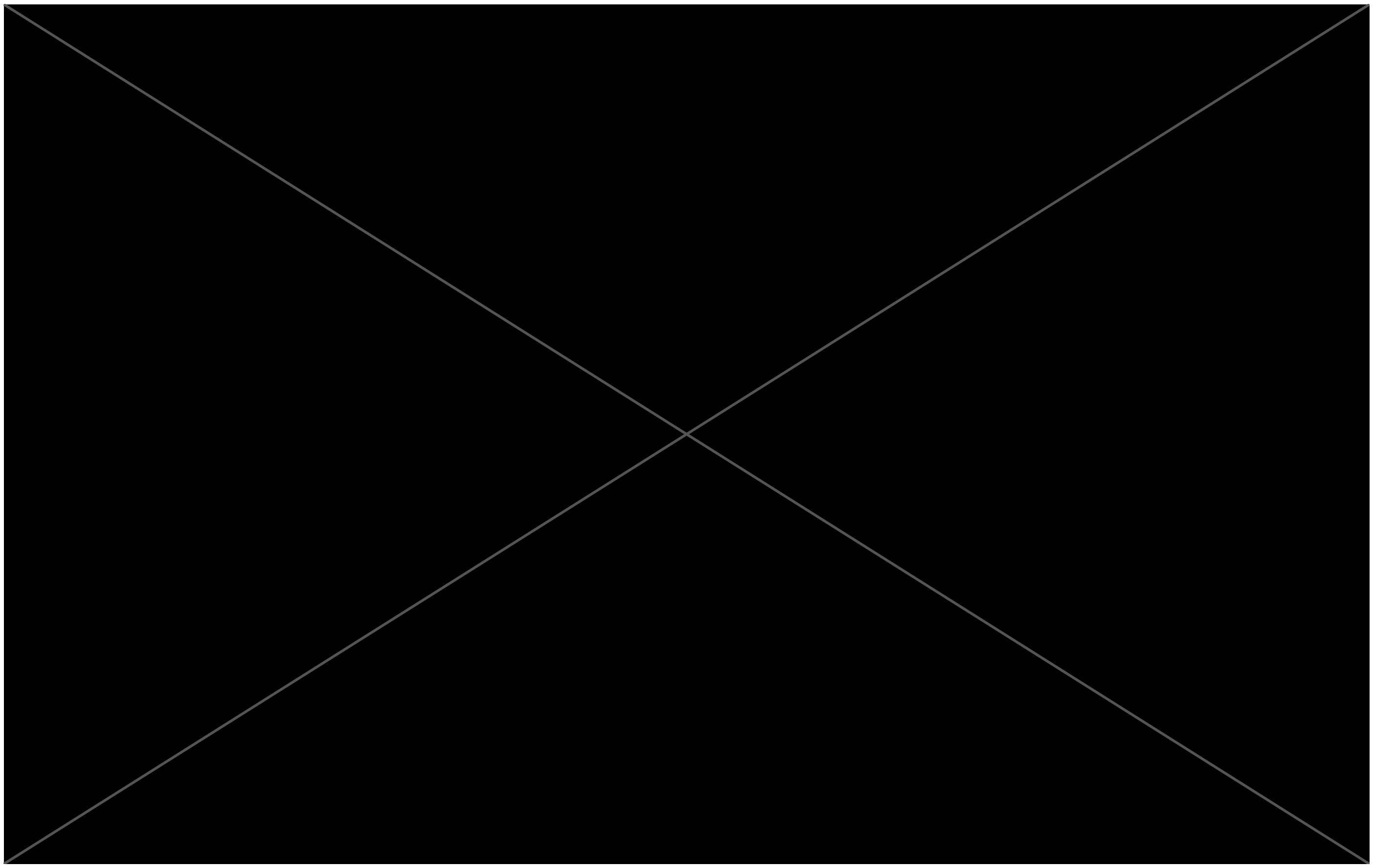
Furthermore, the data underrepresents novel trial designs, such as adaptive protocols. Future work may benefit from natural language processing techniques to extract such design features from unstructured text.

Extrinsic factors

A key limitation of this project is the lack of multiple data sources, capturing external, contextual factors known to mediate trial duration. Factors such as real-time patient recruitment competition, site-specific capacity, and regional patient pool availability were not captured in the dataset, thereby constraining the model's ability to generalise across diverse trial contexts.

At a broader level, further external variables are also unaccounted for challenging to capture, including the evolving regulatory landscape, macroeconomic trends, as well as unprecedented events (e.g. COVID-19), all of which can materially impact recruitment feasibility and operational timelines.

Additionally, only relying on one trial registry may introduce reporting bias. while ClinicalTrials.gov—may introduce geographic or reporting bias.While ClinicalTrials.gov is a prominent trial database covering the majority of trials conducted across the globe, inclusion of other registries, such as European Medicines Agency, 2025(EudraCT) or Australian New Zealand Clinical Trials Registry, 2025 (ANZCTR), could further improve coverage. Similarly, integrating CRO-level operational data or external indicators may enhance predictive accuracies as well.



5 Conclusion

This project resulted in the development of a prototype trial duration prediction tool aimed at augmenting feasibility assessments within clinical operations, [redacted] digital transformation strategy. Using RMSE as the primary evaluation metric, the model met the predefined success criterion for Phase 1 healthy volunteer (P1HV) trials, while models for P1, P2, and P3 trials did not meet performance thresholds. Additionally, R^2 values remained below 50%, underscoring the limited explanatory power of current models and the need for further refinement.

Looking ahead, expanding the dataset to include additional sources, such as CRO databases and external registries, may help capture critical operational variables currently missing from static registries. There is also scope to improve model performance through more advanced feature engineering and the exploration of neural network architectures and multimodal modelling approaches.

However, in light of competing business priorities, further investment in the tool will require a clear demonstration of potential return on investment. Planned feedback collection from the clinical operations team will be key to informing next steps, in alignment with the AGILE methodology applied throughout the project.

Beyond technical outcomes, this project has offered a valuable opportunity for both professional and personal development. It has enhanced my capabilities in machine learning, project scoping, and stakeholder communication, while also reinforcing critical soft skills such as prioritisation and iterative development. [redacted]

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Appendices

Code snippets

```python
import openpyxl as openpyxl
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from datetime import datetime
import re
from xgboost import XGBRegressor
from catboost import CatBoostRegressor
from lightgbm import LGBMRegressor
from scipy.stats import zscore
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.cluster import KMeans
from sklearn.pipeline import Pipeline, make_pipeline
from sklearn.compose import ColumnTransformer
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
    , BayesianRidge
from sklearn.model_selection import train_test_split, RandomizedSearchCV,
    validation_curve, learning_curve
from sklearn.metrics import mean_squared_error, r2_score,
    davies_bouldin_score
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
import numpy as np
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
import os
from sklearn.svm import SVR
from sklearn.neighbors import KNeighborsRegressor
import ast
from sklearn.neural_network import MLPRegressor
```

Listing 1: Libraries used

```python
### Feature engineering: P1 trials (category) (efficacy, safety/tolerability, pharmacology)
columns_to_check = ['Study Title', 'Brief Summary', 'Primary Outcome Measures']
## Function to categorize based on type (efficacy, safety/tolerability, pharmacology)
def categorize_column(text):
    categories = []
    if pd.isna(text):
        return 'none'
    if 'efficacy' in text.lower():
        categories.append('efficacy')
    if 'safety' in text.lower() or 'tolerability' in text.lower():
        categories.append('safety/tolerability')
    if 'pharmaco' in text.lower():
        categories.append('pharmaco')
    # Join the categories with commas if multiple categories are found
    return ', '.join(categories) if categories else 'none'

valid_categories = ['efficacy', 'safety/tolerability', 'pharmaco']

## Function to determine the final category
def determine_final_category(row):
    if row['Phases'] != 1:  # Only process when 'Phases' == 1
        return 'NA'
    final_categories = set()
    # Check the Brief Summary category
    if pd.notna(row['Brief Summary_category']) and row['Brief Summary_category'] != '':
        final_categories.update(row['Brief Summary_category'].replace('and', '').split(', '))
    # Check Primary Outcome Measures if Brief Summary does not contribute categories
    if pd.notna(row['Primary Outcome Measures_category']) and row['Primary Outcome Measures_category'] != '':
        final_categories.update(row['Primary Outcome Measures_category'].replace('and', '').split(', '))
    # Check Study Title column
    if pd.notna(row['Study Title_category']) and row['Study Title_category'] != '':
        final_categories.update(row['Study Title_category'].replace('and', '').split(', '))
    final_categories = [cat.strip() for cat in final_categories if cat.strip() in valid_categories]
    return ', '.join(final_categories) if final_categories else 'none'
```

Listing 2: Part of data preprocessing (Feature engineering: subcategory of P1 trials — by type (efficacy, safety/tolerability, pharmacology))

```python
### Feature engineering: P1 trials (category) (SAD/MAD)
## Function to categorize studies based on SAD and MAD
def categorize_sad_mad(row):
    if row['Phases'] != 1:  # Only process when 'Phases' == 1
        return 'NA'
    categories = []
    # Check for "SAD" and "MAD" in Brief Summary
    if isinstance(row['Brief Summary'], str):
        if 'single' in row['Brief Summary'].lower():
            categories.append('SAD')
        if 'multiple' in row['Brief Summary'].lower():
            categories.append('MAD')
    # Check for "SAD" and "MAD" in Primary Outcome Measures
    if isinstance(row['Primary Outcome Measures'], str):
        if 'single' in row['Primary Outcome Measures'].lower() and 'SAD' not in categories:
            categories.append('SAD')
        if 'multiple' in row['Primary Outcome Measures'].lower() and 'MAD' not in categories:
            categories.append('MAD')
    # Check for "SAD" and "MAD" in Study Title
    if isinstance(row['Study Title'], str):
        if 'single' in row['Study Title'].lower() and 'SAD' not in categories:
            categories.append('SAD')
        if 'multiple' in row['Study Title'].lower() and 'MAD' not in categories:
            categories.append('MAD')
    # Create the final category based on the identified categories
    if 'SAD' in categories and 'MAD' in categories:
        final_category = 'SAD and MAD'  # Both present
    elif categories:
        final_category = ', '.join(set(categories))
    else:
        final_category = 'none'
    return final_category
```

Listing 3: Part of data preprocessing (Feature engineering: subcategory of P1 trials – by dosing design (SAD, MAD))

```python
extended_keyword_dict = {
    'Neoplasms': [
        'cancer', 'melanoma',
        'tumor', 'tumour', 'carcinoma', 'nsclc', 'crc', 'solid malignancies', 'advanced malignancies',
        'hcc', 'neoplasm','Mesothelioma','oncology','carcinoid'
    ],
    'Nervous System Diseases': [
        'alzheimer', 'parkinson','huntington', 'ataxia', 'encephalitis','stroke', 'epilepsy', 'migraine', 'adhd', 'attention deficit','attention-deficit','Hypoactive Sexual Desire Disorder','seizure', 'Wilson disease','tinnitus','Dravet Syndrome','Excessive Sleepiness','sleep disorder',
        'multiple sclerosis', 'amyotrophic lateral sclerosis','epilep', 'spasm', 'glioma', 'brain', 'neuralgia', 'glioblastoma', 'analgesic', "Cerebral palsy", "Delirium", "Dementia", "Faecal incontinence", "Headaches", "Metastatic spinal cord compression", "Motor neurone disease", "Parkinson's disease", "Tremor", "Dystonia", "Spasticity", "Spinal", "Transient ischaemic attack", "Transient loss of consciousness", "Urinary incontinence", 'Neuropathic Pain', 'neuropathy',
        'schizophrenia', 'bipolar', 'analgesia','anxiety', 'depression', 'depressive disorder', 'cognition', 'psycho', 'phobia', 'panic', 'seasonal affective disorder', 'insomnia', 'hypersomnia','narcolepsy','autism','myasthenia gravis','Binge Eating Disorder', "PTSD", "Posttraumatic Stress Disorder", 'post traumatic stress disorder', "cognitive disorders", 'mental disorder','autistic','Mild Cognitive Impairment','borderline personality disorder'
    ],
    'Immune System Diseases': [
        'rheumatoid arthritis', 'rheumatoid', 'lupus', 'spondylitis', 'gout', 'Chronic Spontaneous Urticaria','Graft Versus Host Disease','GVHD','Guillain-Barr  Syndrome', 'colitis', 'inflamma', 'ibd', 'inflammatory bowel disease', 'celiac', 'allergy', 'hay fever','allergic','Type I Hypersensitivity','Sjogren','Shigellosis','restless leg'
    ],
    'Cardiovascular Diseases': [
        'heart failure', 'coronary artery', 'atrial fibrillation', 'coronary', 'cardiomyopathy','vascular disease', 'hypertension', 'cardiac', 'atherosclerosis', 'angina', 'Peripheral Arterial Occlusive Disease', 'arrhythmia', 'myocardial', 'artery', 'arrhythm','cardiovascular','Critical Limb Ischemia'
    ],
    'Nutritional and Metabolic Diseases': [
        'diabetes', 'obesity', 'hyperlipidemia', 'metabolic', 'Nonalcoholic Steatohepatitis', 'chronic kidney disease', 'cholesterol', 'prediabetes', 'diabetes mellitus', "Diabetes, type 1", "NASH",'Dyslipid','NAFLD'
    ],
    'Infectious Diseases': [
        'hiv', 'aids', 'tuberculosis', 'virus', 'infecti', 'Meningococcal', 'Pneumococcal Disease','Invasive Aspergillosis','Herpes','Septic Shock',
        'sepsis', 'covid-19', 'covid', 'influenza', 'viral', 'malaria', 'bacterial', 'fungal','SARS-CoV-2', 'Varicella', 'dengue fever','Vulvovaginal Candidiasis'
    ],
    'Endocrine System Diseases': [
```

```
        'thyroid', 'hypothyroidism', 'hyperthyroidism', 'endocrine', 'Male
    Hypogonadism', 'adrenal', 'pituitary', 'hormone', 'menopause', 'Addison'
    , 'Cushing', 'Graves', 'Hashimoto thyroiditis', 'prolactinoma', 'gland'
    ],
    'Digestive System Diseases': [
        'irritable bowel', 'ibs', 'Irritable Colon', 'gerd', 'gastritis','
    gastroesophageal', 'cirrhosis', 'gastric ulcer', 'reflux disease','
    eosophagitis', 'cholangitis','esophagitis','digestive system disorder',
    'liver', 'hepatitis', 'crohn', 'ulcerative colitis', 'peptic','Erosive
    Esophagitis','functional constipation', 'Constipation - Functional','
    chronic idiopathic constipation', 'biliary','hepatic failure','hepatic
    impairment'
    ],
    'Hemic and Lymphatic Diseases': [
        'leukemia','leukaemia', 'lymphoma', 'anaemia', 'myelodysplastic
    syndrome','MDS','Idiopathic Thrombocytopenic Purpura', 'anemia', '
    hemophilia', 'haemophilia', 'blood', 'thrombocytopenia', 'leukocytosis',
     'Paroxysmal Nocturnal Hemoglobinuria', 'Purpura, Thrombocytopaenic,
    Idiopathic', 'sickle cell', 'thalassemia', 'hematologic', '
    mylodysplastic', 'mpn', 'coagul', 'myeloma', 'neutropenia', 'bleeding',
        "Acute lymphocytic leukemia", "AML","Acute myelogenous leukemia", "
    Amyloidosis", "Antiphospholipid syndrome", "Aplastic anemia", "
    Autoimmune hemolytic anemias", "Autoimmune thrombocytopenia", "Benign
    hematologic conditions", "Bleeding disorders", "Cancer", "Chronic
    lymphocytic leukemia", "Chronic myelogenous leukemia", "Chronic
    myelomonocytic leukemia", "Cryoglobulinemia", "Cutaneous T-cell lymphoma
    ", "Essential thrombocythemia", "Hemoglobinopathies", "Hemophilia", "
    Hereditary hemolytic anemias", "Hodgkin lymphoma", "Immune
    thrombocytopenia", "Iron deficiency anemia", "Lymphoma", "
    Macroglobulinemia", "Monoclonal gammopathy of undetermined significance
    (MGUS)", "Multiple myeloma", "Myelodysplastic syndromes", "Myelofibrosis
    ", "Myeloproliferative disorders", "Natural killer cell leukemia", "Non-
    Hodgkin's lymphoma", "Osteosclerotic myeloma", "Pediatric
    thrombocytopenia", "Pediatric white blood cell disorders", "Polycythemia
     vera", "Sickle cell anemia", "Thalassemia", "Thrombocytopenia", "
    Thrombocytosis", "Vitamin deficiency anemia", "White blood cell
    disorders"
    ],
    'Respiratory Tract Diseases': [
        'asthma', 'copd', 'pulmonary', 'respiratory', 'sinusitis','
    bronchitis','PAH',
        'fibrosis', 'lung', 'bronchiectasis', 'sarcoidosis','Obstructive
    Sleep Apnea', 'pharyngitis'
    ],
    'Male Urogenital Diseases': [
        'benign prostatic hyperplasia','benign prostatic hypertrophy','
    Benign Prostate Hyperplasia', 'bph','Peyronie','Primary Hyperoxaluria','
    Interstitial Cystitis',
        'overactive bladder', 'renal impairment', 'nephropathy', 'chronic
    renal','kidney disease', 'renal disease', 'renal failure'
    ],
    'Eye Diseases': [
        'glaucoma', 'macular', 'retinopathy', 'vision', 'ophthalmic', '
    conjunctivitis','keratitis',
        'dry eye', 'retinitis', "Amblyopia", "Astigmatism", "Blepharitis",
    "Blepharospasm", "Cataracts", "Retinal Vein Occlusion (RVO)", "Visual
```

```
Impairment", "Color Blindness", "Convergence Insufficiency", "Corneal",
"Dry Eye", "Hyperopia", "Floaters", "Glaucoma", "Eye Disease", "
Idiopathic Intracranial Hypertension", "Vision","Macular","Myopia","
Ocular", "Pink Eye", "Presbyopia", "Refractive Errors", "Retinal
Detachment", "Vitreous Detachment"
 ],
 'Skin and Connective Tissue Diseases': [
     'dermatitis', 'eczema', 'psoriasis', 'skin', 'acne', 'Actinic
keratosis', 'Glabellar Lines', 'head lice', 'Pediculosis', 'pruritus', '
Systemic Sclerosis', "bullosa",'leg ulcer','urticaria',
     'vitiligo', 'alopecia', 'rosacea', "dermato", 'diabetic foot', '
pressure ulcer', 'skin infection', 'Seborrheic Keratosis','Prurigo
Nodularis','scleroderma', 'derma'
 ],
 'Congenital, Hereditary, and Neonatal Diseases and Abnormalities': [
     'Congenital Bleeding Disorder|Haemophilia A','Pompe Disease', '
cystic fibrosis', 'duchenne', 'gaucher', 'rare disease', 'orphan', '
Primary Immunodeficiency', 'congenital','X-linked Hypophosphatemia','
Alpha1-antitrypsin Deficiency', 'muscular dystrophy', 'phenylketonuria',
 "Adrenoleukodystrophy", "Gaucher disease", "Glucose galactose
malabsorption", "Hereditary hemochromatosis", "Lesch-Nyhan syndrome", "
Maple syrup urine disease", "Menkes syndrome", "Niemann-Pick disease", "
Phenylketonuria", "Prader-Willi syndrome", "Porphyria", "Refsum disease"
, "Tangier disease", "Tay-Sachs disease", "Wilson's disease", "Zellweger
 syndrome", "Castleman disease", "Fanconi anemia", "Hairy cell leukemia"
, "Hypereosinophilic syndrome", "Large granular lymphocyte disorders", "
POEMS syndrome", "Systemic capillary leak syndrome", "Systemic
mastocytosis", "Von Willebrand disease", "Waldenstrom macroglobulinemia"
, "Bietti's Crystalline Dystrophy", "Coloboma", "Retinitis Pigmentosa",
"Stargardt Disease", "Usher Syndrome"
 ],
 'Musculoskeletal Diseases': [
     "Arthritis", "Fractures", "Hip conditions", "Joint replacement", "
Knee conditions", "Maxillofacial conditions", "Osteoporosis", "Spinal
conditions",'Degenerative Disc Disease','Osteopenia'
 ],
 'Chemically-Induced Disorders': [
     'substance abuse', 'depend','drug induced','drug-induced','
addiction', 'opioid', 'opioid-induced', 'cocaine', 'Drug Interaction
Potentiation','Nausea and Vomiting, Chemotherapy-Induced','smoking','
Opioid-induced Constipation','Alcohol Use Disorder','drug abuse'
 ],
 'Female Urogenital Diseases and Pregnancy Complications': [
     'contraception','Uterine Fibroids','Vaginal Atrophy','female
infertility'
 ],
 'Pathological Conditions, Signs and Symptoms': [
     'Sedation','dyspepsia','postoperative pain','post operative pain','
post-operative','postsurgical pain', 'lower back pain', 'low back pain',
'pain, acute', 'lower-back pain','scar'
 ],
 'Stomatognathic Diseases': [
     'Oral Mucositis','Periodontal Disease'
 ],
 'Wound and Injuries': [
     'wound','Ankle Injuries'
```

```
    ],
    'Otorhinolaryngologic Diseases': [
        'Meniere','Nasal Congestion','Nasal Polyposis'
    ]
}
```

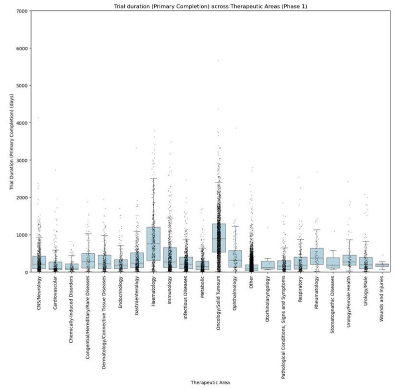
Listing 4: Dictionary for mapping trials to therapeutic areas

```python
region_keyword_dict = {
    'US': ['united states', 'usa', 'california', 'texas', 'new york', 'boston', 'chicago', 'florida', 'miami', 'los angeles', 'san francisco', 'washington', 'dc', 'seattle', 'atlanta', 'dallas'],
    'North America (excluding US)': ['canada', 'toronto', 'vancouver', 'montreal', 'ottawa'],
    'Europe': ['germany', 'france', 'italy', 'spain', 'united kingdom', 'uk', 'england', 'scotland', 'wales', 'ireland', 'netherlands', 'belgium', 'sweden', 'denmark', 'norway', 'finland', 'poland', 'greece', 'portugal', 'switzerland', 'austria', 'hungary', 'romania', 'czech republic', 'europe'],
    'Asia': ['china', 'japan', 'india', 'south korea', 'singapore', 'taiwan', 'hong kong', 'thailand', 'malaysia', 'philippines', 'indonesia', 'vietnam', 'pakistan', 'bangladesh', 'sri lanka', 'nepal', 'mongolia', 'myanmar'],
    'Australia/Oceania': ['australia', 'sydney', 'melbourne', 'brisbane', 'perth', 'adelaide', 'new zealand', 'fiji', 'papua new guinea', 'samoa', 'tonga', 'solomon islands', 'vanuatu'],
    'Latin America': ['brazil', 'argentina', 'chile', 'colombia', 'peru', 'venezuela', 'uruguay', 'paraguay', 'ecuador', 'bolivia', 'guyana', 'suriname', 'mexico', 'guatemala', 'honduras', 'el salvador', 'nicaragua', 'costa rica', 'panama', 'belize'],
    'MENA': ['israel', 'saudi arabia', 'uae', 'united arab emirates', 'qatar', 'lebanon', 'oman', 'kuwait', 'bahrain', 'jordan', 'iraq', 'iran', 'palestine', 'syria', 'yemen', 'egypt', 'morocco', 'algeria', 'tunisia', 'libya'],
    'Sub-Saharan Africa': ['south africa', 'nigeria', 'kenya', 'ethiopia', 'ghana', 'uganda', 'tanzania', 'zambia', 'zimbabwe', 'angola', 'ivory coast', 'senegal', 'sudan']
}
```

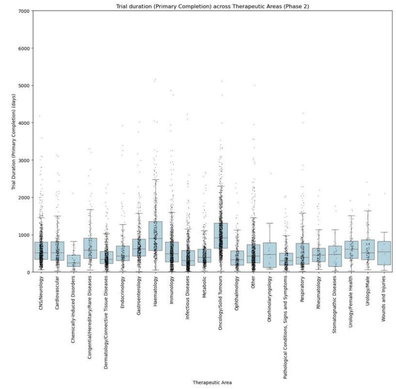
Listing 5: Dictionary for mapping trials to geographical regions

Figures and Tables

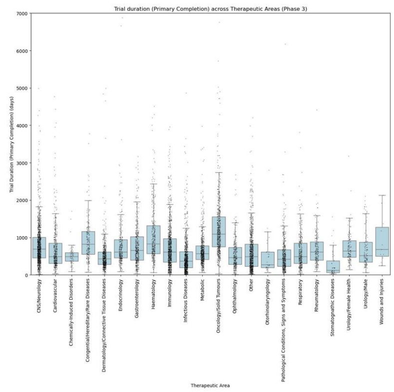
Figures below show the pattern of variance in trial durations across therapeutic areas (Supplementary Fig 1 and 2) and regions (Supplementary Fig 3 and 4) for each trial phase.



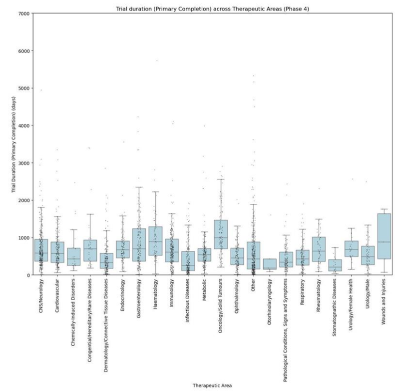
(a) Phase 1 trials



(b) Phase 2 trials

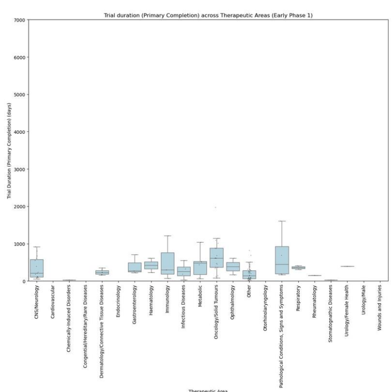


(c) Phase 3 trials

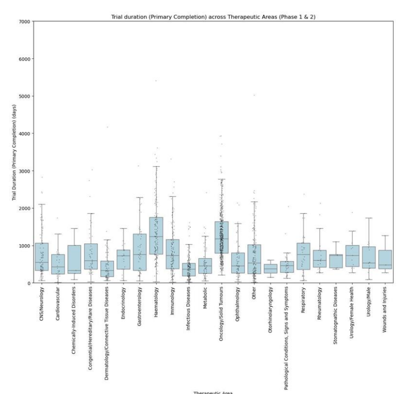


(d) Phase 4 trials

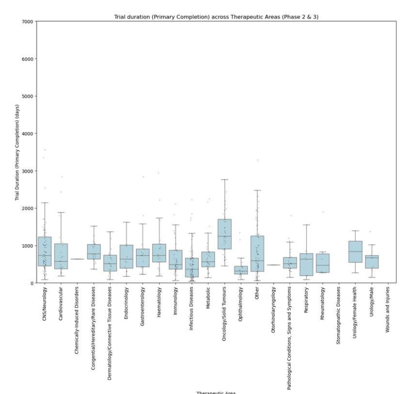
Supplementary Figure 1. Trial duration variances across therapeutic areas for the 4 phase-specific trial datasets. Similar patterns of trial durations are observed across therapeutic areas.



(a) Early Phase 1 trials

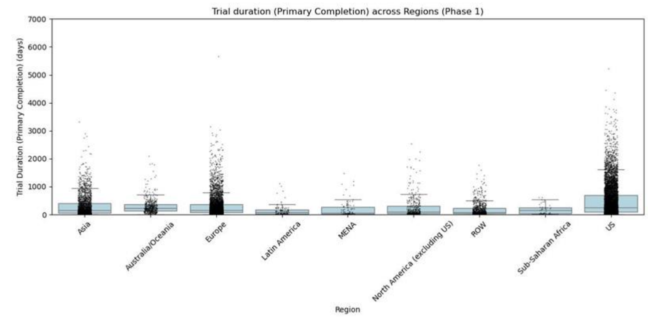


(b) Cross-over Phase 1/2 trials

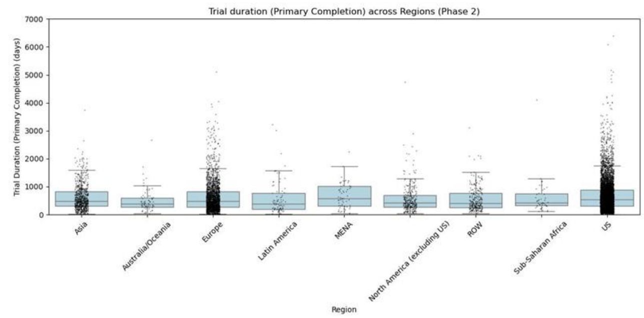


(c) Cross-over Phase 2/3 trials

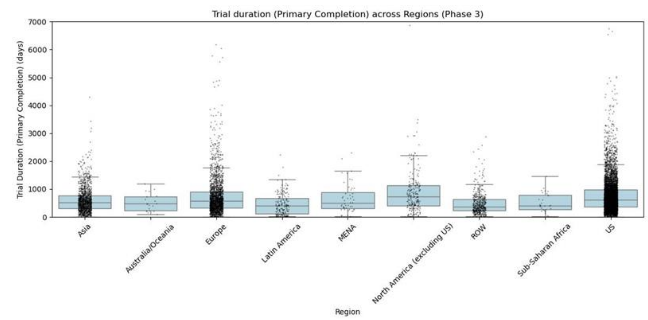
Supplementary Figure 2. Few data points and high variance/noise observed in trial duration across therapeutic areas for Early Phase 1 as well as cross-over P1/2 and P2/3 trials. This serves as justification for the exclusion of these trials from the selected phase-specific datasets for model training purposes.



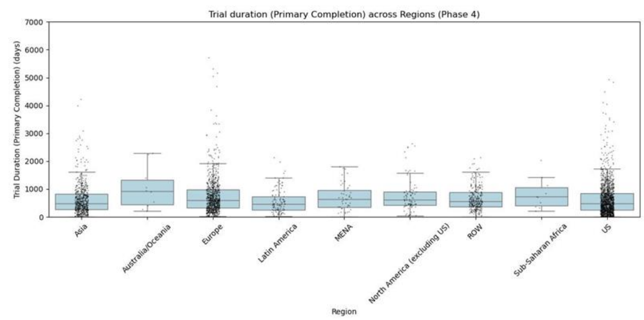
(a) Phase 1 trials



(b) Phase 2 trials

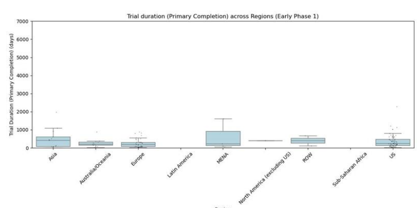


(c) Phase 3 trials

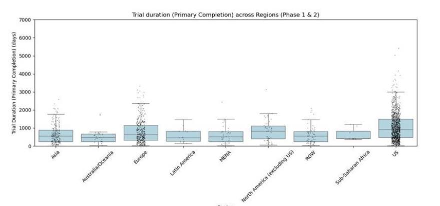


(d) Phase 4 trials

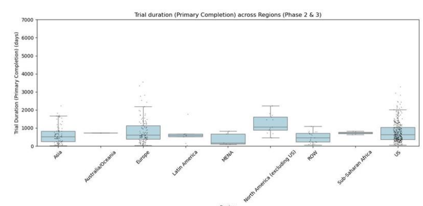
Supplementary Figure 3. Trial duration variances across regions for the 4 phase-specific trial datasets. Similar patterns of trial durations are observed across regions.



(a) Early Phase 1 trials



(b) Cross-over Phase 1/2 trials



(c) Cross-over Phase 2/3 trials

Supplementary Figure 4. Few data points and high variance/noise observed in trial duration across regions for Early Phase 1 as well as cross-over P1/2 and P2/3 trials. This serves as justification for the exclusion of these trials from the selected phase-specific datasets for model training purposes.